MBY4CJ 203. ENVIRONMENTAL AND SANITATION MICROBIOLOGY

Programme	B. Sc. Microbiology				
Course Code	MBY4CJ 203				
Course Title	Environmental and Sanitation Microbiology				
Type of Course	Major				
Semester	IV				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Nil				
Course Summary	This course explores the role of microorganisms in environmental and sanitation contexts, focusing on their impact on air and water quality, waste management, and their use in bioremediation and pollution control.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand microbial dynamics in air and their implications on health.	(U)	(C)	Internal Exam, End Semester Exam
CO2	Evaluate methods of microbial sampling and monitoring in environmental settings.	(E)	(P)	Practical Assessments, Assignments
CO3	Analyze aquatic ecosystems and the role of microbes in water quality management.	(An)	(C)	Assignments, End Semester Exam
CO4	Apply microbial techniques for solid waste management and bioremediation.	(Ap)	(P)	Practical Assessments
CO5	Discuss the impact of microbial processes on pollution control and environmental restoration.	(U)	(C)	Internal Exam, Assignments

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
\# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)

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Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks -70
I	**Microbiology of air**		**10**	**15**
	1	Atmospheric layers, organisms in air, distribution and sources		
	2	Disease forecasting in plants		
	3	Indoor and outdoor air		
	4	Droplet nuclei, aerosol, infectious dust		
	5	Microbiological sampling of air - gravity slide, plate exposure, vertical cylinder, Hirst spore trap, Rotorod sampler, Andersen sampler, hand held air sampler, impingers and filtration. Advantages and disadvantages of these techniques		
	6	Brief account of air borne transmission of harmful microbes and air borne infections		
II	**Aquatic Microbiology**		**10**	**15**
	7	Aquatic environment, distribution of microorganisms in aquatic environment - fresh water, estuarine and marine water systems		
	8	Factors influencing growth and distributions		
	9	Water Purification procedures for single dwelling and municipal water supplies		
	10	Concept of indicator organisms, Microbiological examination of water. BOD, COD		
	11	Wastewater treatment steps and methods		
	12	Eutrophication and algal bloom		
	13	Brief account of water borne diseases and transmission		
III	**Solid Waste Management**		**15**	**25**
	14	Sources and types of solid waste		
	15	need for management		
	16	Landfills, composting, vermi- composting, anaerobic digesters, methanogenesis and production of biogas		
	17	Design and management of biogas plants		
IV	**Bioremediation**		**10**	**15**
	18	Novel pollutants, persistence and biomagnification		

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	19	Recalcitrant halocarbons- nitroaromatic compounds, PCB, alkyl benzene sulphonates		
	20	Petroleum hydrocarbons - their biodegradation		
	21	Bioremediation of polluted environment - Oil spills, heavy Metals and other xenobiotics.		
	22	Microbial leaching and corrosion of metals		
V	**Open ended**			
	1	Marine Natural products from marine microorganisms- antibiotics, toxins, organic acids, biosurfactants, pigments, biopolymers and enzymes		
	2	Waste management strategies in the local bodies: Discussion, Visit, evaluation, suggestion for improvements		

Books and References:

1. Textbook of Biochemistry by Lehninger
2. Biochemistry by Stryer
3. Molecular Biology of the Gene by Watson, JD, Hopkins NH, Roberts JW, Steitz JA,
4. Weiner AAM, 1987. The Benjamin/Cummings publishing company.
5. Genes V by Lewin B, 1994. Oxford University Press.
6. Molecular Cell Biology by Lodish, H, Baltimore D, Berk A, Zipursky SL, Matsudaira P,
7. Darnell J., 1995. Scientific American Books.
8. Molecular Biology by Freifelder D., 1991 Narosa Publishing Home.
9. Principles of Gene Manipulation, 4th Ed., by R.S.Old and S.B.Primrose. 1989. Blackwell Scientific Publications, London
10. Cell Biology by Karp

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Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2		1			3	2	1		2	1
CO2	2	3	3	2			2	1	3	2	3	2
CO3	1	2	3		2	1	2	3	2	1	1	3
CO4	3		2	3	1	2	1	2	3	3	2	1
CO5	2	1		2	3		3	2	1	2	3	

Correlation Levels: 1-Slightly / Low, 2-Moderate / Medium, 3-Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Continuous Assessment (10%)
- Practical Assessment (20%)
- Endsemester Exam (70%)

Mapping of COs to Assessment Rubrics :

CO	Internal Exam	Assignment	End Semester Examination	Practical Assessment
CO1	✓		✓	
CO2		✓		✓
CO3		✓	✓	
CO4				✓
CO5	✓	✓		

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PYQ Question

Topics: Atmospheric layers, sampling devices, indoor/outdoor microflora, and airborne infections.

2-Mark Questions

1. **Define Droplet Nuclei:** Small particles (1–5 μm) formed by the evaporation of moisture from respiratory droplets; they can remain suspended in air for long periods.
   What is an Impinger? A sampling device that collects microorganisms by bubbling air through a liquid medium.
   Gravity Slide: A passive air sampling method where particles settle onto a sticky slide due to gravity.
2. **Define Aerosol:** A suspension of fine solid particles or liquid droplets in a gaseous medium like air.
3. **Hirst Spore Trap:** A suction device used for continuous monitoring of airborne spores over a period of time.

5-Mark Questions

1. **Explain the principles and working of the Anderson Sampler:** It is a multi-stage sieve sampler that separates particles based on size, simulating the human respiratory system.

2. **Summarize the differences between indoor and outdoor air microflora:** Outdoor air is influenced by soil and water sources, while indoor air is dominated by human shedding and ventilation efficiency.

3. **Inspect major air-borne infections with suitable examples:** Discusses transmission of viral (H1N1, Influenza) and bacterial (Tuberculosis) diseases through air.

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10+ Mark Questions

1. **Investigate the various methods and devices used for the microbiological quality testing of air:** Detailed essay on impaction, impingement, and filtration methods.

Module II: Aquatic Microbiology

Topics: Freshwater/Marine ecosystems, water purification, indicator organisms, and BOD/COD.

2-Mark Questions

1. **Indicator Organisms:** Microbes whose presence indicates potential fecal contamination, such as *E. coli* or Coliforms.

2. **BOD (Biochemical Oxygen Demand):** The amount of dissolved oxygen needed by aerobic biological organisms to break down organic material in water.

3. **Eutrophication:** The process where water bodies receive excess nutrients that stimulate excessive plant/algal growth.

4. **Benthic Zone:** The lowest ecological region of a body of water, including the sediment surface.

5. **Trickling Filter:** A biological treatment process that uses a fixed bed of media to treat wastewater.

5-Mark Questions

1. **Formulate the steps involved in the purification of municipal water:** Describes the sequence of screening, coagulation, sedimentation, filtration, and disinfection.

2. **Explain the application of membrane filter techniques in water quality studies:** Discusses how water is passed through a thin filter to trap and count bacteria.

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Module III: Solid Waste Management

Topics: Landfills, composting, anaerobic digestion, and biogas production.

2-Mark Questions

1. **Methanogenesis:** The formation of methane by microbes known as methanogens under anaerobic conditions.

2. **Vermi-composting:** A process using earthworms to convert organic waste into high-quality manure.

3. **Anaerobic Digester:** An airtight tank where bacteria break down organic material in the absence of oxygen.

4. **Composition of Biogas:** Primarily consists of Methane (CH_4) and Carbon Dioxide (CO_2).

5-Mark Questions

1. **Elucidate the design and management of a biogas plant:** Describes the digester, gas holder, and the process of loading organic feedstock.

2. **Discuss the principles and applications of vermi-composting:** Explains how earthworms like *Eisenia fetida* speed up the decomposition of organic waste.

Module IV: Bioremediation

Topics: Xenobiotics, oil spills, bioleaching, and biocorrosion.

2-Mark Questions

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1. **Recalcitrant:** Substances that are resistant to microbial degradation and persist in the environment.

2. **Biomagnification:** The process by which toxins become more concentrated in living tissues as they move up the food chain.

3. **Bioleaching:** The use of microorganisms to extract metals from their ores.

4. **Superbug:** A genetically engineered microorganism (like *Pseudomonas putida*) capable of degrading complex pollutants.

10+ Mark Questions

1. **Outline the applications of genetically engineered bacteria for the bioremediation of oil spills:** Discusses the role of specialized microbes in breaking down petroleum hydrocarbons.

2. **Examine the major steps involved in the treatment of sewage:** A comprehensive review of primary, secondary (biological), and tertiary treatment.

Part A: Short Answer (2 Marks)

1. **Define Bio-scrubbers:** Devices used for removing pollutants from air using microbial biofilms.
2. **What is a Biosurfactant?** Surface-active substances produced by microbes to emulsify oil during bioremediation.
3. **Role of *Eisenia fetida*:** The specific species of earthworm most commonly used in vermi-composting.
4. **Distinguish between Bioaugmentation and Biostimulation:** Adding foreign microbes vs. adding nutrients to stimulate existing microbes.
5. **What are "Novel Pollutants"?** Emerging environmental contaminants like microplastics or pharmaceutical waste.

Part B: Short Essay (5 Marks)

1. **Discuss the "Windrow" vs. "Static Pile" Composting Methods:** Compare the manual turning of waste versus passive aeration in solid waste management.
2. **Explain the significance of the Redfield Ratio in aquatic systems:** Discuss the ratio of Carbon, Nitrogen, and Phosphorus (106:16:1) and its impact on algal blooms.
3. **Describe the role of Thiobacillus in biocorrosion of pipelines:** Explain how sulfur-oxidizing bacteria produce sulfuric acid that degrades metal.
4. **Outline the advantages of Biological over Chemical treatment in sewage plants:** Focus on cost-effectiveness and the reduction of toxic sludge.

Part C: Essay (10+ Marks)

1. **Investigate the impact of Hydrostatic Pressure and Temperature on marine microflora:** Discuss how piezophilic and psychrophilic bacteria survive in deep-sea estuarine and marine systems.
2. **Design a comprehensive plan for Solid Waste Management in a local urban body:** Use your knowledge of landfills, anaerobic digesters, and vermi-composting to propose a waste-to-energy model.
3. **Critically analyze the persistence of PCBs and Halocarbons in the environment:** Discuss the chemical structures that make them recalcitrant and the specific microbial pathways required for their degradation

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